

MODULE 01 / 40

Northern Cardinal

*Cardinalis cardinalis***QUICK STATS**

Length	8.3 – 9.3 in (21 – 23.5 cm)
Wingspan	9.8 – 12.2 in (25 – 31 cm)
Weight	1.5 – 1.7 oz (42 – 48 g)
Lifespan	Up to 15 years (avg. 3 in wild)

PRESENTATION

The Northern Cardinal is one of the most iconic and instantly recognizable birds visiting North American gardens year-round. The brilliant scarlet male, with his dramatic crest and orange-red bill, is among the most photographed birds on the continent. The female, dressed in warm buff-brown with reddish tinges on her crest, wings and tail, is no less elegant. Both sexes remain on their territory throughout winter, bringing color and song to snow-covered gardens when most other species have migrated south.

HABITAT & RANGE

The Northern Cardinal thrives across the eastern half of North America, from southern Maine and the Great Lakes south through Florida, and westward to the Great Plains of Texas and Oklahoma. Populations are also well-established in Arizona, New Mexico, and parts of California. Southern Ontario and Quebec host significant breeding populations. The species has expanded its range northward over the past century, likely aided by the proliferation of bird feeders and milder winters. It favors woodland edges, dense shrubby thickets, hedgerows, suburban gardens, and parks with mature trees and dense understory cover.

DIET & FEEDING

- Black-oil sunflower seeds — the undisputed favorite
- Safflower seeds — especially attractive to cardinals
- White millet and cracked corn
- Wild berries: dogwood, holly, elderberry, sumac
- Insects and larvae (essential protein during breeding season)
- Fruit: grapes, mulberries, raspberries

BEHAVIOR & SOCIAL LIFE

Cardinals are territorial songbirds that maintain year-round home ranges of roughly 2–4 acres. Males fiercely defend their territory through song and, famously, by attacking their own reflection in windows — mistaking it for a rival. They are monogamous and pairs often remain together through multiple breeding seasons. Outside the breeding season, cardinals may gather in loose flocks of up to 70 individuals at abundant food sources. Males frequently feed females during courtship — a behavior called mate-feeding — passing seeds beak-to-beak.

NESTING & BREEDING

Breeding season runs from March through September, with females typically raising 2 to 4 broods per year — one of the highest rates among North American songbirds. The female constructs a cup-shaped nest over 3–9 days using twigs, bark strips, grasses and leaves, lined with fine grasses and hair. Nests are built 1–15 feet above ground in dense shrubs, vines or low trees. Each clutch contains 2–5 eggs (usually 3), pale whitish or greenish with brown speckles. Incubation lasts 11–13 days, performed almost exclusively by the female. Nestlings fledge at 9–11 days and the male takes over their care while the female begins a new nest.

SONG & COMMUNICATION

The Northern Cardinal is renowned for its rich, loud whistled song — a series of clear slurred notes often described as "cheer-cheer-cheer" or "birdy-birdy-birdy." Both males and females sing, which is unusual among North American songbirds. The female often sings from the nest, possibly communicating food needs to her mate. Cardinals have a repertoire of 16 or more distinct song patterns. Their sharp metallic "chip" call is an alarm note used to warn of predators. Song peaks in late winter and early spring as males establish territories.

CONSERVATION STATUS

IUCN Red List:	Least Concern	Population trend:	Stable / Increasing
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With an estimated population of over 100 million individuals, the Northern Cardinal is one of the most abundant songbirds in North America. Its range has expanded northward significantly since the 1900s. It is the official state bird of 7 US states — more than any other species.

FEEDER TIPS

- Best feeder type: hopper feeder or large platform feeder
- Preferred seeds: safflower (squirrels tend to avoid it) and black-oil sunflower
- Cardinals feed most actively at dawn and dusk
- Place feeders near dense shrubs — cardinals prefer cover close by
- Offer a shallow birdbath: cardinals bathe and drink regularly

COLORING GUIDE

ZONE	RECOMMENDED COLOR
Male body	Vivid red — cadmium red
Facial mask	Deep black
Bill	Orange-coral
Female body	Warm sandy brown
Wings & crest ■	Rusty-red highlights
Eye	Dark brown, black ring

COLOR ME!

Northern Cardinal • *Cardinalis cardinalis* • Module 01



MODULE 02 / 40

Black-capped Chickadee

*Poecile atricapillus***QUICK STATS**

Length	4.7 – 5.9 in (12 – 15 cm)
Wingspan	6.3 – 8.3 in (16 – 21 cm)
Weight	0.3 – 0.5 oz (9 – 14 g)
Lifespan	Up to 12 years (avg. 2–3 in wild)

PRESENTATION

The Black-capped Chickadee is one of the most beloved and recognizable backyard birds across northern North America. Bold, curious, and remarkably tolerant of humans, it is often the first species to discover a newly installed feeder. Its cheerful "chick-a-dee-dee-dee" call is one of the most familiar sounds of North American winters. Despite its tiny size, the chickadee is supremely adapted to cold climates, capable of surviving temperatures well below freezing through a combination of extraordinary memory, metabolic flexibility, and dense winter plumage.

HABITAT & RANGE

The Black-capped Chickadee occupies a vast range across the northern tier of North America, from Alaska and the Yukon east to Newfoundland and Nova Scotia, and south through the northern United States including New England, the Great Lakes region, and the Rocky Mountain states down to northern New Mexico. It inhabits deciduous and mixed forests, forest edges, riparian corridors, suburban parks, and wooded gardens. It is a permanent resident throughout its range and does not migrate, relying instead on cached food and physiological adaptations to endure harsh winters.

DIET & FEEDING

- Black-oil sunflower seeds — the top choice at any feeder
- Suet cakes — critical high-energy winter food source
- Nyjer (thistle) seed and hulled sunflower chips
- Peanut pieces and peanut butter mixtures
- Mealworms (live or dried) during breeding season
- Wild: insects, insect eggs, spiders, berries, seeds

BEHAVIOR & SOCIAL LIFE

Chickadees are highly social birds that form stable winter flocks of 6–12 individuals with a clear dominance hierarchy. These flocks often serve as the nucleus for mixed-species foraging groups, attracting nuthatches, woodpeckers, and warblers. Within the flock, dominant birds feed first and claim the best roost sites. The chickadee's "chick-a-dee" alarm call is remarkably sophisticated: the number of "dee" notes appended to the call signals the level of threat, with more notes indicating a more dangerous predator nearby. Researchers have identified it as one of the most complex vocalizations in the animal kingdom.

NESTING & BREEDING

Breeding pairs form in late winter as flocks begin to break up. Chickadees are cavity nesters, excavating their own nest holes in soft rotting wood or using natural cavities and nest boxes. The female lines the cavity with moss, plant fibers, fur, and feathers, creating a deep insulating cup. Clutches typically contain 6–8 white eggs with reddish-brown speckles. Incubation lasts 12–13 days, performed by the female alone while the male brings her food. Both parents feed the nestlings, which fledge at 16 days. Pairs raise one brood per year.

SONG & COMMUNICATION

The chickadee has two primary vocalizations. Its contact and alarm call — the iconic "chick-a-dee-dee-dee" — varies in complexity based on context and threat level. Its territorial song is a clear, whistled two-note "fee-bee" (or "fee-bee-bee"), the first note higher than the second, heard most often on mild late-winter days as males begin to establish breeding territories. Chickadees also produce a variety of soft gargling calls within their flocks.

CONSERVATION STATUS

IUCN Red List:	Least Concern	Population trend:	Stable
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The Black-capped Chickadee remains abundant across its vast range, with an estimated population exceeding 100 million individuals. It adapts well to suburban and urban environments and benefits from nest box programs and winter feeding.

FEEDER TIPS

- Tube feeders or small hopper feeders work best
- Black-oil sunflower seeds are strongly preferred over striped
- Chickadees can be hand-tamed with patience over several weeks
- Place a nest box (1 1/8" hole diameter) near woodland edge
- Suet feeders are especially valued from October through March

COLORING GUIDE

ZONE	RECOMMENDED COLOR
Crown & bib	Deep black
Cheek patches	Pure white
Back & wings	Slate gray
Belly	White to pale gray
Flanks	Buffy-white
Bill	Short, black

COLOR ME!

Black-capped Chickadee • *Poecile atricapillus* • Module 02



MODULE 03 / 40

Blue Jay

*Cyanocitta cristata***QUICK STATS**

Length	9.8 – 11.8 in (25 – 30 cm)
Wingspan	13.4 – 16.9 in (34 – 43 cm)
Weight	2.5 – 3.5 oz (70 – 100 g)
Lifespan	Up to 26 years (avg. 7 in wild)

PRESENTATION

The Blue Jay is one of the most striking and intelligent birds to visit North American gardens. Its bold blue, white, and black plumage makes it instantly recognizable at any feeder. Despite its reputation as a bully — it dominates most other species at feeding stations — the Blue Jay plays a vital ecological role as one of the continent's most important seed dispersers. Highly adaptable and extraordinarily vocal, it produces a wide variety of calls including convincing imitations of hawks, which it uses to clear feeders of competing birds.

HABITAT & RANGE

The Blue Jay is found throughout the eastern half of North America, from southern Canada (Alberta east to Nova Scotia and Newfoundland) south through Florida and west to the edge of the Great Plains. It inhabits a wide range of wooded environments including deciduous and mixed forests, forest edges, suburban neighborhoods, city parks, and gardens with mature oak trees. Northern populations are partially migratory, with large flocks moving south in autumn along coastlines and ridges, though many individuals remain year-round even in cold climates.

DIET & FEEDING

- Acorns — by far the most important food source
- Whole and shelled peanuts — a feeder favorite
- Sunflower seeds (black-oil and striped)
- Corn (cracked or whole kernel)
- Insects, grasshoppers, beetles during summer
- Occasionally: eggs and nestlings of other birds

BEHAVIOR & SOCIAL LIFE

Blue Jays are highly social outside the breeding season, gathering in loose flocks and traveling together through the forest canopy. They are among the most vocal birds in North America, producing a rich repertoire of calls including their signature loud "jay-jay-jay" alarm, musical liquid gurgles, whisper songs, and remarkably accurate mimicry of Red-shouldered and Red-tailed Hawks. This mimicry serves multiple purposes: alerting other jays to hawk presence and, controversially, potentially clearing feeders so the jay can feed undisturbed. Blue Jays have demonstrated problem-solving abilities and tool use in laboratory settings, ranking among the most cognitively advanced birds studied.

NESTING & BREEDING

Blue Jays form monogamous pairs that often mate for life. Both sexes participate in nest building, constructing a bulky open cup of twigs, bark strips, mud, and grass in the fork of a tree or shrub, typically 10–25 feet above ground. Clutches of 2–7 eggs (usually 4–5) are incubated for 17–18 days by both parents. Nestlings are fed by both parents and fledged at 17–21 days. Pairs typically raise one brood per year. Both parents aggressively mob and chase away predators, including owls, hawks, cats, and humans.

SONG & COMMUNICATION

The Blue Jay's vocal repertoire is one of the most varied of any North American bird. Its primary call is a loud, harsh "jay" or "jeer," used as an alarm. It also produces a bell-like "tool-tool" call, soft musical whisper songs audible only at close range, and uncannily accurate imitations of at least three hawk species. Researchers believe jays use hawk mimicry both as a warning system and as a competitive strategy at feeders.

CONSERVATION STATUS

IUCN Red List:	Least Concern	Population trend:	Stable
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Blue Jay populations are stable across their range, with an estimated 45 million individuals in North America. The species has benefited from suburban expansion and the planting of oak trees in urban environments.

FEEDER TIPS

- Platform feeders or large hopper feeders accommodate their size
- Whole peanuts in the shell are irresistible and provide great viewing
- Scatter cracked corn on the ground — jays readily feed at ground level
- Plant oak trees: a single oak can produce thousands of acorns annually
- Expect jays to cache food — they may visit and depart rapidly with full beaks

COLORING GUIDE

ZONE	RECOMMENDED COLOR
Crest & upper wings	Bright blue with black bars
Tail	Blue with black barring, white tips
Face & nape	White with bold black necklace
Breast & belly	White to pale gray
Back	Blue-gray
Bill & legs	Black

COLOR ME!

Blue Jay • *Cyanocitta cristata* • Module 03



MODULE 04 / 40

House Finch

*Haemorhous mexicanus***QUICK STATS**

Length	5.0 – 6.0 in (12.7 – 15.2 cm)
Wingspan	8.0 – 10.0 in (20.3 – 25.4 cm)
Weight	0.6 – 0.9 oz (16 – 27 g)
Lifespan	Up to 11 years (avg. 2–3 in wild)

PRESENTATION

The House Finch is one of the most abundant and widespread songbirds in North America, found from coast to coast in virtually every habitat where humans have settled. Originally native to the western United States and Mexico, it was introduced to the eastern US in the 1940s when illegal pet trade birds were released on Long Island, New York. It has since colonized the entire continent with remarkable speed. The male's rosy-red head and breast make it one of the most colorful regular visitors to backyard feeders, while the streaked brown female is a master of camouflage.

HABITAT & RANGE

Today the House Finch is found across all 48 contiguous United States, southern Canada, and Mexico. It thrives in urban and suburban environments, farmlands, forest edges, and desert scrub. Highly adaptable, it nests on building ledges, in ornamental conifers, hanging baskets, and even inside cacti in the Southwest. Unlike many finches, it does not undertake long migrations, remaining year-round across most of its range.

DIET & FEEDING

- Nyjer (thistle) seed — strongly preferred
- Black-oil sunflower seeds and hulled chips
- White millet scattered on platform feeders
- Wild berries: mulberry, pyracantha, crabapple
- Flower buds and plant shoots in spring
- Weed seeds from dandelion, chickweed, and thistle

BEHAVIOR & SOCIAL LIFE

House Finches are highly gregarious birds that gather in flocks of dozens or even hundreds outside the breeding season. They are cheerful, active, and vocal at feeders, often tolerating close human approach. Males sing a long, rambling warble year-round — one of the few North American songbirds that sings regularly through winter. Fascinatingly, the intensity of the male's red coloration is entirely diet-dependent: birds consuming more carotenoid-rich foods (berries, certain plants) develop brighter red plumage, while poorly-fed males may appear orange or yellow. Females actively choose the reddest males as mates, using color as a reliable indicator of foraging ability and health.

NESTING & BREEDING

House Finches are remarkably flexible nesters, using an extraordinary range of sites including tree cavities, cliff ledges, building eaves, hanging flower baskets, and old nests of other species. The female builds a compact cup of grasses, leaves, and fine plant fibers. Clutches of 2–6 pale blue eggs with dark speckles are incubated for 13–14 days by the female alone. The male feeds her during incubation. Nestlings fledge at 12–19 days. Pairs raise 2–3 broods per year, sometimes beginning a new nest before the previous brood has fully fledged.

SONG & COMMUNICATION

The male House Finch produces one of the most melodious and sustained songs of any common feeder bird — a rich, cheerful warble lasting 3 seconds or more, delivered from a prominent perch. Song is heard year-round but peaks in spring. The call is a sharp "cheep" often given in flight. Flocks maintain contact with soft chirping notes while foraging together.

CONSERVATION STATUS

IUCN Red List:	Least Concern	Population trend:	Stable
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With an estimated population of over 267 million individuals, the House Finch is one of the most numerous birds in North America. However, populations in the eastern US suffered dramatic declines in the 1990s due to a contagious eye disease (Mycoplasmal conjunctivitis) that spread rapidly through feeder populations. Numbers have since partially recovered. If you see a finch with red, swollen, or crusty eyes at your feeder, clean feeders immediately with a 10% bleach solution.

FEEDER TIPS

- Tube feeders with small ports are ideal for finches
- Nyjer seed in a dedicated thistle sock attracts large flocks
- Clean feeders regularly — House Finches are susceptible to eye disease
- Platform feeders with millet attract ground-feeding individuals
- House Finches often feed in large, noisy groups — expect crowds

COLORING GUIDE

ZONE	RECOMMENDED COLOR
Head & breast (male)	Raspberry red to orange-red
Back & wings	Brown with dark streaking
Belly	White with brown streaks
Rump (male)	Reddish-pink
Female (entire)	Plain brown, heavily streaked
Bill	Short, conical, gray-brown

COLOR ME!

House Finch • *Haemorhous mexicanus* • Module 04



MODULE 05 / 40

Ruby-throated Hummingbird

*Archilochus colubris***QUICK STATS**

Length	2.8 – 3.5 in (7 – 9 cm)
Wingspan	3.1 – 4.3 in (8 – 11 cm)
Weight	0.07 – 0.21 oz (2 – 6 g)
Lifespan	Up to 9 years (avg. 3–5 in wild)

PRESENTATION

The Ruby-throated Hummingbird is the only hummingbird species that regularly breeds in eastern North America, and one of the most extraordinary birds on the continent. Weighing less than a nickel, it is a marvel of biological engineering: its wings beat 53 times per second, its heart beats up to 1,200 times per minute during flight, and it is the only bird capable of sustained backwards flight. The male's iridescent ruby-red gorget (throat patch) can appear jet black in poor lighting, then flash brilliant red as the angle of light changes — one of nature's most spectacular optical effects.

HABITAT & RANGE

During summer, the Ruby-throated Hummingbird breeds across the entire eastern half of North America, from Nova Scotia and Manitoba south through Florida and west to the Great Plains. It inhabits forest edges, gardens, orchards, and meadows with abundant flowering plants. Each fall, it undertakes one of the most astonishing migrations in the bird world: a non-stop flight of 500–900 miles across the Gulf of Mexico to its wintering grounds in Mexico and Central America. It returns north each spring from April through May.

DIET & FEEDING

- Flower nectar — trumpetvine, salvia, monarda, impatiens
- Feeder solution: 1 part white sugar to 4 parts water (no dye)
- Tiny insects and spiders — essential protein source
- Tree sap from sapsucker wells (used opportunistically)
- Never use honey, artificial sweeteners, or red dye in feeders

BEHAVIOR & SOCIAL LIFE

Ruby-throated Hummingbirds are among the most aggressively territorial birds in North America relative to their size. Males fiercely defend feeding territories, dive-bombing intruders — including birds many times their size — with surprising ferocity. Outside of courtship and mating, males and females live entirely solitary lives. To survive cold nights and food shortages, hummingbirds enter a state of torpor: their body temperature drops from 40°C to around 18°C, their heart rate falls from 1,200 to just 50 beats per minute, and their breathing nearly stops. This metabolic shutdown can save up to 60% of their nightly energy expenditure.

NESTING & BREEDING

The female alone builds one of nature's most remarkable nests: a walnut-sized cup of plant down and spider silk, camouflaged on the outside with lichens, and expandable as the chicks grow. It is typically placed on a slender downward-sloping branch 10–20 feet above ground. She lays exactly 2 white eggs the size of peas, incubating them alone for 11–14 days. She feeds the nestlings by inserting her bill deep into their throats and regurgitating nectar and insects. Young fledge at 18–22 days. The male plays no role in nesting or chick-rearing.

SONG & COMMUNICATION

Ruby-throated Hummingbirds are not known for melodious song. Their primary vocalizations are sharp, rapid chips and chattering calls used during territorial disputes and pursuit chases. Males perform spectacular U-shaped or pendulum dive displays during courtship, producing a distinctive wing-buzz at the bottom of each arc. The wingbeat itself produces the characteristic humming sound that gives the family its name.

CONSERVATION STATUS

IUCN Red List:	Least Concern	Population trend:	Stable
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With a stable population estimated at nearly 34 million individuals, the Ruby-throated Hummingbird is not currently threatened. However, it faces pressure from habitat loss on its wintering grounds in Central America and from climate change affecting the timing of flower blooms relative to migration. Planting native nectar plants and maintaining clean feeders are the most impactful ways to support local populations.

FEEDER TIPS

- Use a red feeder — no need to add dye to the solution
- Recipe: boil 1 cup white sugar in 4 cups water, cool completely
- Change solution every 2–3 days in hot weather to prevent fermentation
- Clean feeders with hot water and a bottle brush — no soap needed
- Hang feeders by late April in the South, mid-May in the North
- Plant trumpet vine, bee balm, salvia, and cardinal flower nearby

COLORING GUIDE

ZONE	RECOMMENDED COLOR
Back & crown (both)	Iridescent emerald green
Gorget — male throat	Ruby red (may look black)
Breast & belly	White to pale gray
Tail (male)	Forked, dark green-black
Female throat	White, no red gorget
Bill	Long, straight, black

COLOR ME!

Ruby-throated Hummingbird • *Archilochus colubris* • Module 05



MODULE 06 / 40

American Robin

*Turdus migratorius***QUICK STATS**

Length	7.9 – 11.0 in (20 – 28 cm)
Wingspan	12.2 – 15.7 in (31 – 40 cm)
Weight	2.7 – 3.0 oz (77 – 85 g)
Lifespan	Up to 14 years (avg. 2 in wild)

PRESENTATION

The American Robin is one of the most familiar and beloved birds in North America — a true symbol of spring whose return each year is eagerly anticipated across the continent. Plump, upright, and boldly patterned with its signature orange-red breast, it is a consummate garden bird, foraging confidently across lawns for earthworms and nesting on ledges, trees, and even window sills in close proximity to humans. Despite being called a robin, it is actually a thrush — a member of the same family as bluebirds and solitaires — and is the largest thrush in North America.

HABITAT & RANGE

The American Robin breeds across virtually all of North America, from Alaska and northern Canada south through Mexico, making it one of the continent's most widespread birds. It inhabits an extraordinary range of environments: lawns, parks, forests, mountains up to the treeline, tundra edges, and urban centers. Northern populations migrate south in autumn, though many winter in the southern US where berry-laden trees provide food. Year-round residents are common throughout much of the contiguous United States. Large winter flocks of thousands can gather in roosts.

DIET & FEEDING

- Earthworms — located by sight and sound, not smell
- Wild berries: holly, crabapple, juniper, dogwood, cedar
- Insects, beetles, grasshoppers, and caterpillars
- Fruit: grapes, cherries, plums, serviceberry
- Rarely visits seed feeders — prefers open ground foraging
- Plant berry-producing shrubs to attract robins to your garden

BEHAVIOR & SOCIAL LIFE

American Robins are highly visible and active birds, spending much of their time foraging on the ground with a characteristic run-stop-tilt posture as they listen and watch for earthworms. Contrary to popular belief, they locate worms primarily by sight rather than hearing, though both senses play a role. Outside breeding season, robins are highly gregarious, forming communal roosts of thousands or even millions of birds in woodlands. These roosts disperse each morning as birds fan out to forage, then reconvene at dusk. Males arrive on breeding grounds before females each spring and immediately begin singing to establish territories.

NESTING & BREEDING

The American Robin is one of the earliest nesters in North America, often beginning as soon as late February in the South. The female builds a robust cup nest of grasses, twigs, and worm castings, reinforced with mud and lined with fine grasses. Nests are placed on building ledges, in tree forks, or on fence posts 5–15 feet above ground. The female lays 3–5 eggs of a vivid, unmistakable sky blue — the origin of the color name "robin's egg blue." Incubation lasts 12–14 days. Both parents feed nestlings, which fledge at 13 days. Pairs typically raise 2–3 broods per season.

SONG & COMMUNICATION

The robin's song is one of the most recognized sounds of the North American spring — a rich, melodious caroling of rising and falling phrases often transcribed as "cheerily, cheer-up, cheerio." Males begin singing before dawn and continue into dusk, sometimes even singing after dark near artificial lights. The sharp "tut-tut-tut" alarm call warns of ground predators, while a high thin "see" alerts to aerial threats. Robins are among the first birds to sing in the morning chorus.

CONSERVATION STATUS

IUCN Red List:	Least Concern	Population trend:	Stable
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With an estimated population of 300 million individuals, the American Robin is one of the most abundant birds in North America. Its adaptability to human-modified landscapes has helped it thrive despite widespread habitat changes. It does face localized threats from pesticide use, which reduces earthworm availability and can cause direct poisoning.

FEEDER TIPS

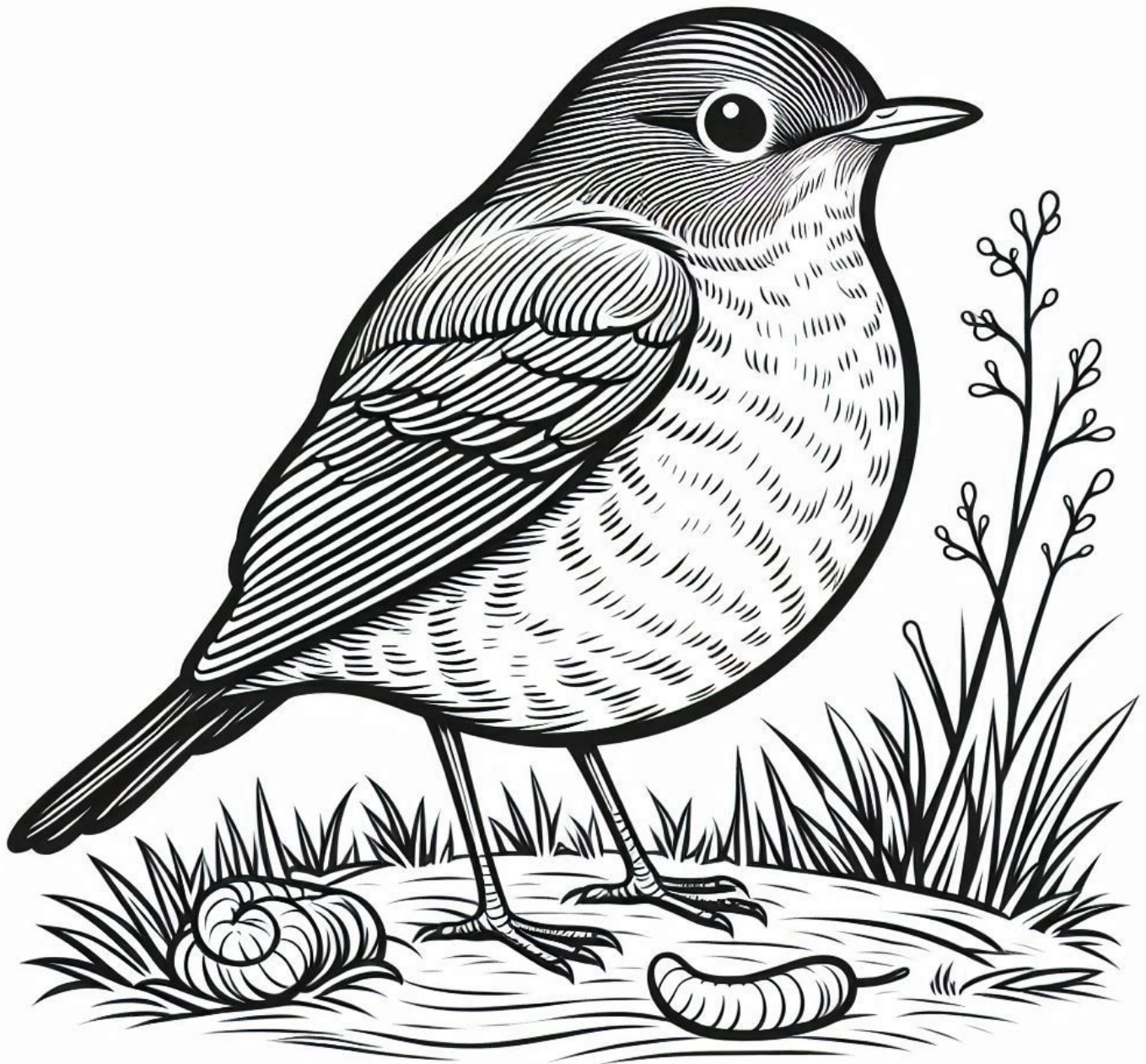
- Robins rarely use seed feeders — attract them differently
- Install a shallow birdbath: robins love to bathe and drink
- Plant holly, crabapple, dogwood, or serviceberry for winter berries
- Avoid pesticides on lawns — robins depend on healthy earthworm populations
- A platform feeder with mealworms or fruit pieces may attract them

COLORING GUIDE

ZONE	RECOMMENDED COLOR
Head & back	Dark gray to black
Breast & belly	Brick orange-red
Throat	White with dark streaks
White eye-ring	Pure white crescent
Bill	Yellow-orange
Legs & feet	Dark brown-gray

COLOR ME!

American Robin • *Turdus migratorius* • Module 06



MODULE 07 / 40

Downy Woodpecker

Dryobates pubescens

QUICK STATS

Length	5.5 – 6.7 in (14 – 17 cm)
Wingspan	9.8 – 11.8 in (25 – 30 cm)
Weight	0.7 – 1.0 oz (20 – 33 g)
Lifespan	Up to 11 years (avg. 1–2 in wild)

PRESENTATION

The Downy Woodpecker is the smallest woodpecker in North America and one of the most familiar visitors to backyard feeders across the continent. Despite its diminutive size, it is a bold and active bird, clinging to suet cages, sunflower feeders, and tree trunks with equal ease. Its crisp black-and-white plumage and the male's bright red nape patch make it easy to identify. The Downy is often confused with the larger Hairy Woodpecker, but can be distinguished by its much shorter bill relative to head size and its smaller overall dimensions. It is one of the most widespread woodpeckers in North America and a year-round resident throughout its range.

HABITAT & RANGE

The Downy Woodpecker inhabits a remarkable range of forested and semi-open environments across North America, from Alaska and northern Canada south through most of the United States, absent only from the arid Southwest and parts of the Great Plains. It thrives in deciduous and mixed forests, orchards, riparian corridors, suburban parks, and gardens with mature trees. Unlike many woodpeckers, it readily adapts to fragmented and urban habitats and is a common year-round feeder visitor across most of its range.

DIET & FEEDING

- Suet cakes — the single most effective food to offer
- Black-oil sunflower seeds and hulled chips
- Peanut butter (plain, no additives) smeared on bark
- Bark insects: beetle larvae, ants, moth cocoons
- Wild berries and plant galls in winter
- Poison ivy berries — an important winter food source

BEHAVIOR & SOCIAL LIFE

Downy Woodpeckers are active, acrobatic birds that forage across a wide variety of surfaces — tree trunks, large branches, weed stems, and even corn stalks. Their small size allows them to exploit foraging niches unavailable to larger woodpeckers, including the tips of small branches and slender plant stems where insect eggs and larvae overwinter. Outside breeding season, Downies frequently join mixed-species foraging flocks with chickadees, nuthatches, and kinglets, benefiting from the collective vigilance of the group. Males and females partition foraging habitat even within a territory: males tend to forage on smaller branches and weed stems, females on larger trunks.

NESTING & BREEDING

Downy Woodpeckers excavate their own nest cavities in dead or dying trees, a process that takes both sexes 1–3 weeks to complete. The entrance hole is perfectly round, about 1.25 inches in diameter, leading to a gourd-shaped chamber 6–12 inches deep lined only with wood chips. No nest material is added. Clutches of 3–8 white eggs are incubated by both parents for 12 days. The male incubates at night. Nestlings are fed by both parents and fledge at 20–25 days. Old nest cavities are vital for other cavity-nesting species including chickadees, bluebirds, and small owls.

SONG & COMMUNICATION

The Downy Woodpecker's primary call is a sharp, flat "pik" — shorter and higher-pitched than the similar Hairy Woodpecker's call. It also produces a descending whinny call used during territorial interactions and a rattling "drum" — a rapid series of bill strikes on a resonant dead branch — used to establish territory and attract mates. Drumming rates average about 17 strikes per second. Both sexes drum, though males drum more frequently.

CONSERVATION STATUS

IUCN Red List:	Least Concern	Population trend:	Stable
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The Downy Woodpecker population is estimated at over 14 million individuals and remains stable across its range. It benefits from dead tree retention in managed forests and suburban areas. Its excavated nest cavities provide essential housing for dozens of other cavity-dependent species, making it a keystone species in many woodland ecosystems.

FEEDER TIPS

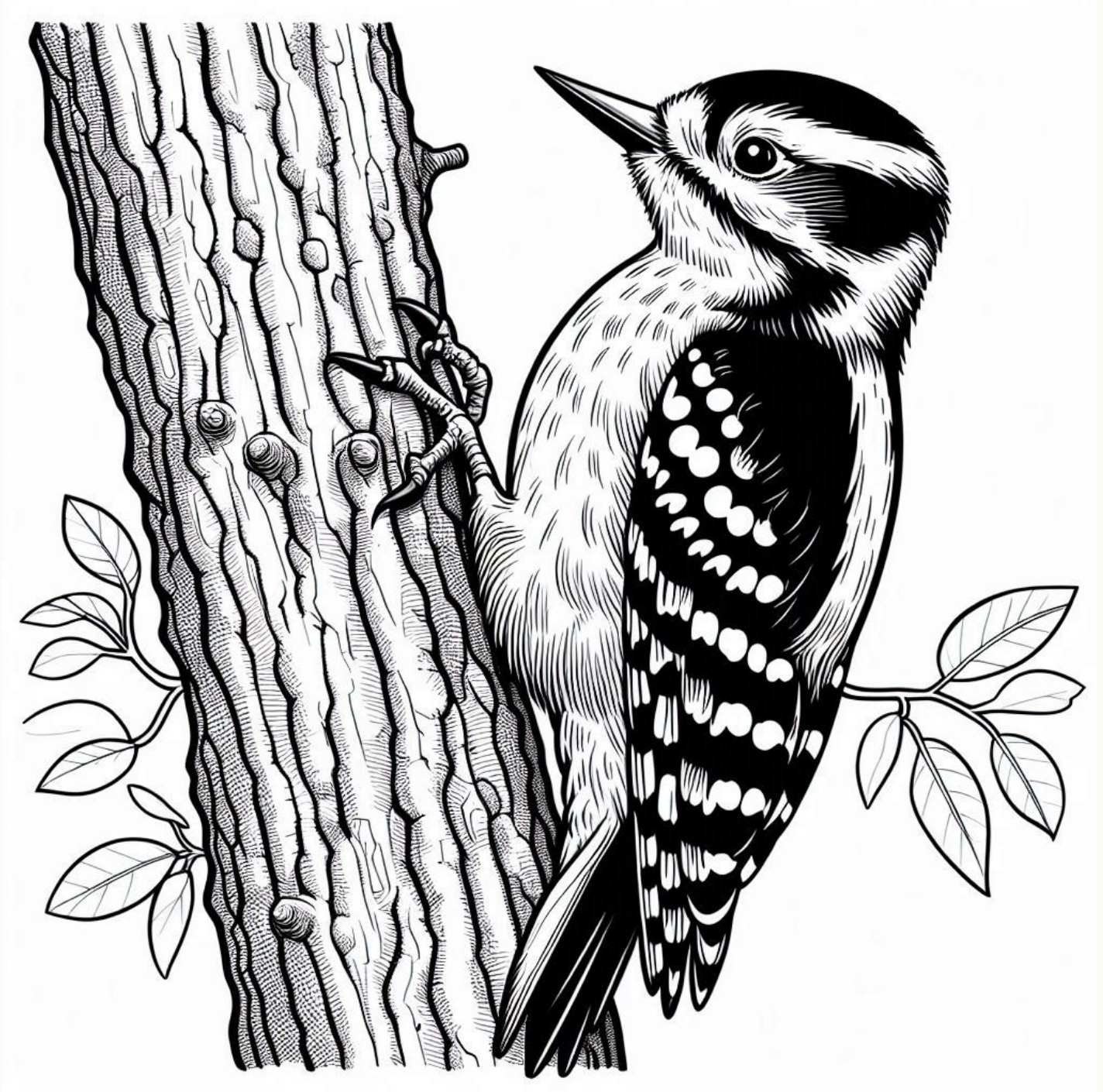
- Suet cages are the most reliable way to attract Downy Woodpeckers
- Offer suet year-round — especially critical in cold winters
- Upside-down suet feeders deter starlings while Downies feed easily
- Hang feeders near tree trunks — Downies feel safer close to cover
- Leave dead trees standing when safe — essential for nesting and foraging

COLORING GUIDE

ZONE	RECOMMENDED COLOR
Crown & back	<i>Solid black</i>
Wing spots	<i>Black with white dots/bars</i>
White back stripe	<i>Pure white center stripe</i>
Underparts	<i>White to pale buff</i>
Red nape patch (male)	<i>Bright red — rear of crown only</i>
Bill	<i>Short, dark gray, chisel-shaped</i>

COLOR ME!

Downy Woodpecker • *Dryobates pubescens* • Module 07



MODULE 08 / 40

White-breasted Nuthatch

*Sitta carolinensis***QUICK STATS**

Length	5.1 – 5.5 in (13 – 14 cm)
Wingspan	7.9 – 10.6 in (20 – 27 cm)
Weight	0.6 – 1.1 oz (18 – 30 g)
Lifespan	Up to 10 years (avg. 2 in wild)

PRESENTATION

The White-breasted Nuthatch is one of the most distinctive and entertaining birds to visit backyard feeders — instantly recognizable by its habit of walking headfirst down tree trunks, a behavior unique among North American birds. This remarkable feat is made possible by an unusually large hind toe that provides a powerful grip regardless of orientation. Compact, neckless, and boldly patterned in black, white, and blue-gray, it is a year-round resident that pairs for life and maintains a fixed territory through all seasons. Its loud, nasal calls are among the most recognizable winter sounds in eastern North American forests.

HABITAT & RANGE

The White-breasted Nuthatch is found across most of the United States and southern Canada, from British Columbia east to Nova Scotia and south through the Appalachians and into Mexico. It inhabits mature deciduous and mixed forests, forest edges, orchards, parks, and wooded suburban gardens with large trees. It shows a strong preference for oak, hickory, and beech-dominated forests where its primary wild food — acorns and nuts — is abundant. It is a permanent resident throughout its range and does not migrate.

DIET & FEEDING

- Sunflower seeds — black-oil and striped both accepted
- Suet cakes — especially valued in winter
- Whole and shelled peanuts
- Acorns and hickory nuts (cached in bark crevices)
- Insects and spiders gleaned from bark year-round
- Peanut butter mixtures smeared into bark furrows

BEHAVIOR & SOCIAL LIFE

White-breasted Nuthatches are monogamous and maintain year-round pair bonds, with mates staying in close contact through constant soft calls. They are vigorously territorial, with pairs defending the same woodland territory across multiple years. Their signature head-down foraging posture gives them a unique advantage: they can spot insects and cached food items hidden in bark crevices from angles that upward-foraging birds like woodpeckers would miss entirely. They are avid food cachers, wedging seeds and nuts into bark furrows and covering them with lichen or bark fragments — a behavior that has earned them the folk name "Devil Downhead" in some regions. They often join winter foraging flocks with chickadees and Downy Woodpeckers.

NESTING & BREEDING

Breeding begins in March or April. Pairs use natural tree cavities, old woodpecker holes, or nest boxes, with the female doing most of the nest building. She constructs a cup of bark strips, grasses, and fur inside the cavity. A fascinating defensive behavior: the pair rubs crushed insects (particularly blister beetles) around the entrance hole — the noxious chemicals may deter squirrels and other predators. Clutches of 5–9 white eggs speckled with red-brown are incubated by the female for 13–14 days. Both parents feed the nestlings, which fledge at 26 days — an unusually long nestling period.

SONG & COMMUNICATION

The White-breasted Nuthatch is one of the most vocal winter birds. Its primary call is a loud, nasal "yank yank yank" — a rapid series that carries well through winter woodland. A softer, higher "ip" contact call keeps pairs in touch while foraging. The male's song, heard mostly in late winter and early spring, is a rapid series of low whistled notes "whi-whi-whi-whi" repeated several times. Females also vocalize frequently with soft nasal calls.

CONSERVATION STATUS

IUCN Red List:	Least Concern	Population trend:	Stable
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The White-breasted Nuthatch population is estimated at approximately 10 million individuals and is considered stable. It benefits from the retention of large mature trees with natural cavities, and from nest box programs in managed forests. Forest fragmentation and the loss of old-growth deciduous woodland are the primary long-term concerns for this species.

FEEDER TIPS

- Offer sunflower seeds in a tube or hopper feeder near large trees
- A suet cage mounted on a tree trunk is ideal for nuthatches
- Smear peanut butter into bark crevices — they will find it quickly
- Install a nest box with a 1.25–1.5 inch entrance hole on a large tree
- Nuthatches cache seeds — place feeders near rough-barked oaks or maples

COLORING GUIDE

ZONE	RECOMMENDED COLOR
Crown & nape (male)	Glossy black cap
Crown (female)	Dark gray — not black
Back & wings	Blue-gray
Face & underparts	White to pale
Flanks & undertail	Rusty chestnut-orange
Bill	Long, pointed, dark gray

COLOR ME!

White-breasted Nuthatch • *Sitta carolinensis* • Module 08



MODULE 09 / 40

American Goldfinch

*Spinus tristis***QUICK STATS**

Length	4.3 – 5.1 in (11 – 13 cm)
Wingspan	7.5 – 8.7 in (19 – 22 cm)
Weight	0.4 – 0.7 oz (11 – 20 g)
Lifespan	Up to 11 years (avg. 3–6 in wild)

PRESENTATION

The American Goldfinch is one of the most brilliantly colored birds in North America — the male's summer plumage of blazing canary yellow and jet black is unmistakable at any feeder. Known affectionately as the "wild canary," it is one of the few finches that undergoes a complete molt twice a year, transitioning from dazzling yellow in summer to subdued olive-brown in winter. It is also one of the latest nesters in North America, deliberately timing its breeding to coincide with the peak availability of thistle seeds in late summer — the primary food used to feed its nestlings.

HABITAT & RANGE

The American Goldfinch breeds across southern Canada and the northern two-thirds of the United States, from British Columbia east to Newfoundland and south through the Appalachians. In winter, northern populations migrate south, and the species can then be found across the entire continental US and into northern Mexico. It inhabits open and semi-open environments: weedy fields, meadows, floodplains, roadsides, forest edges, orchards, and suburban gardens. It shows a strong preference for areas with abundant thistles, sunflowers, and other composite flowers.

DIET & FEEDING

- Nyjer (thistle) seed — the top choice, attracts large flocks
- Black-oil sunflower seeds and hulled chips
- Native thistle, coneflower, and sunflower seeds in the wild
- Dandelion, aster, and goldenrod seeds
- Almost entirely vegetarian — rarely eats insects
- Birch and alder catkins in late winter

BEHAVIOR & SOCIAL LIFE

American Goldfinches are highly social birds that travel in flocks year-round, even during the breeding season when most songbirds become territorial. Flocks can number in the hundreds at productive feeding sites. Their flight pattern is distinctively undulating — a series of flaps followed by a glide — and they frequently call in flight with a cheerful "per-chick-o-ree" that announces their presence overhead. Males are loosely territorial around nest sites but tolerate other goldfinches feeding nearby. They are among the most acrobatic feeder birds, clinging upside-down to thistle socks and swinging seed heads with ease.

NESTING & BREEDING

The American Goldfinch is the latest nesting songbird in North America, typically not beginning until late June or July — months after most other species have already fledged young. This timing is not accidental: it synchronizes perfectly with the peak ripening of thistle and other composite seeds. The female weaves an exceptionally tight, compact cup nest of plant fibers and thistle down, bound with spider silk, in the fork of a shrub or small tree. The nest is so tightly woven it can hold water. She lays 4–6 pale blue eggs, incubating them alone for 12–14 days while the male feeds her. Both parents feed nestlings regurgitated seeds. Young fledge at 11–17 days.

SONG & COMMUNICATION

The goldfinch's song is a long, rambling series of musical twitters, warbles, and trills — bright and cheerful, delivered from a perch or in flight. The flight call "per-chick-o-ree" or "po-ta-to-chip" is one of the most recognizable sounds of open country in summer. A soft "swee" contact call keeps flock members in touch. Males sing most actively from late spring through summer.

CONSERVATION STATUS

IUCN Red List:	Least Concern	Population trend:	Declining slightly
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With a population estimated at 42 million individuals, the American Goldfinch remains common but has shown gradual declines in recent decades, linked to loss of weedy meadow habitat, agricultural intensification, and the reduction of native thistles. Planting native wildflowers — especially coneflowers, black-eyed Susans, and native thistles — is one of the most effective ways to support goldfinch populations.

FEEDER TIPS

- Nyjer (thistle) seed in a dedicated thistle sock or tube feeder
- Expect large flocks — install multiple feeders to reduce competition
- Keep nyjer fresh — it goes stale quickly and goldfinches will reject it
- Plant native coneflowers and leave seed heads standing through winter
- Goldfinches arrive later in the day — most active mid-morning to afternoon

COLORING GUIDE

ZONE	RECOMMENDED COLOR
Body (male, summer)	Brilliant canary yellow
Wings & tail	Black with white wing bars
Forehead patch (male)	Black cap — front only
Body (female)	Olive-yellow, duller than male
Body (male, winter)	Olive-brown — loses bright yellow
Bill	Short, conical, orange-pink

COLOR ME!

American Goldfinch • *Spinus tristis* • Module 09



Dark-eyed Junco

Junco hyemalis

QUICK STATS

Length	5.5 – 6.3 in (14 – 16 cm)
Wingspan	7.1 – 9.8 in (18 – 25 cm)
Weight	0.6 – 1.1 oz (18 – 30 g)
Lifespan	Up to 11 years (avg. 3 in wild)

PRESENTATION

The Dark-eyed Junco is one of the most abundant birds in North America and a quintessential winter feeder visitor, earning its nickname "snowbird" for its tendency to appear at feeders just as the first cold weather arrives. Crisp and tidy in its dark gray-and-white plumage, it is among the easiest birds to identify — the sudden flash of white outer tail feathers as it flies away is one of the most recognizable field marks in North American birding. The Dark-eyed Junco is actually a complex of several regional forms that were once considered separate species, including the Slate-colored, Oregon, Pink-sided, and White-winged Juncos.

HABITAT & RANGE

The Dark-eyed Junco breeds across the boreal forests of Canada and Alaska, and in montane coniferous forests south through the Appalachians, Rockies, and Sierra Nevada. In winter it migrates south and can be found across virtually the entire contiguous United States, making it one of the most widespread winter birds on the continent. It favors woodland edges, brushy areas, and open ground near cover. In the breeding season it inhabits mature coniferous and mixed forests, often at higher elevations.

DIET & FEEDING

- White millet — strongly preferred, especially on the ground
- Black-oil sunflower seeds and cracked corn
- Milo, sorghum, and mixed seed blends
- Weed seeds: chickweed, sorrel, buckwheat
- Insects and spiders during breeding season
- Juncos are primarily ground feeders — scatter seed on the ground

BEHAVIOR & SOCIAL LIFE

Dark-eyed Juncos are highly social in winter, forming flocks of 10–30 birds that maintain a clear dominance hierarchy. Males dominate females, and adults dominate young birds. Within a flock, subordinate individuals are pushed to the edges of feeding areas where predator risk is higher. Despite this hierarchy, flock members cooperate in vigilance — when one bird detects a threat and flushes, the flash of white tail feathers from multiple birds simultaneously serves as a visual alarm signal. Juncos are primarily ground feeders, hopping and scratching through leaf litter and snow with both feet simultaneously in a characteristic double-scratch motion.

NESTING & BREEDING

Juncos nest on or near the ground, often tucking their cup nest under an overhanging root, rock ledge, or clump of vegetation on a steep slope or bank. The female builds a neat cup of grasses, moss, and bark strips, lined with fine grasses and hair. Clutches of 3–5 eggs (pale bluish or grayish, spotted with brown) are incubated by the female for 12–13 days. Both parents feed the nestlings, which fledge at just 9–13 days — unusually quickly, likely an adaptation to reduce nest predation risk. Pairs raise 2–3 broods per season.

SONG & COMMUNICATION

The junco's song is a simple, pleasant musical trill on one pitch, similar to a Chipping Sparrow but slower and more musical. It is heard primarily on the breeding grounds. In winter flocks, juncos communicate with a variety of sharp "tick" and "smacking" call notes, as well as soft twittering calls within the flock. The sharp alarm call triggers the characteristic tail-flashing response.

CONSERVATION STATUS

IUCN Red List:	Least Concern	Population trend:	Stable
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With an estimated population of 630 million individuals, the Dark-eyed Junco is one of the most numerous birds in North America. It is estimated that juncos make up roughly 71% of all birds at feeders across the US during winter. Population trends are stable across most of its range.

FEEDER TIPS

- Scatter white millet directly on the ground — juncos prefer ground feeding
- A low platform feeder or ground tray works better than tube feeders
- Clear snow from feeding areas — juncos struggle to dig through deep snow
- Dense shrubs nearby provide essential escape cover from hawks
- Juncos often arrive in mixed flocks with sparrows and chickadees

COLORING GUIDE

ZONE	RECOMMENDED COLOR
Hood & back (male)	<i>Slate gray to dark charcoal</i>
Belly	<i>White — sharp contrast with hood</i>
Outer tail feathers	<i>White — flash when flying</i>
Female hood	<i>Paler brown-gray, less contrasted</i>
Bill	<i>Short, pale pink — conical</i>
Legs	<i>Pale pink</i>

COLOR ME!

Dark-eyed Junco • *Junco hyemalis* • Module 10



MODULE 11 / 40

Song Sparrow

Melospiza melodia

QUICK STATS

Length	4.7 – 6.7 in (12 – 17 cm)
Wingspan	7.1 – 9.4 in (18 – 24 cm)
Weight	0.4 – 1.9 oz (11 – 53 g)
Lifespan	Up to 11 years (avg. 3 in wild)

PRESENTATION

The Song Sparrow is one of the most widespread and variable songbirds in North America, found in nearly every habitat from Alaska to Mexico. Despite its streaky brown plumage that many birders initially dismiss as "just a sparrow," it is a remarkably accomplished singer — the male begins his rich, complex song as early as February, often singing from an exposed perch even in snow and freezing temperatures. The Song Sparrow is also one of the most scientifically studied birds in the world, with research spanning over a century that has illuminated fundamental principles of ecology, behavior, and evolution.

HABITAT & RANGE

The Song Sparrow occupies an extraordinary range of habitats across North America, from the Aleutian Islands of Alaska south through Canada and across the entire continental US to central Mexico. It inhabits wetland edges, marshes, shrubby fields, forest edges, riparian thickets, coastal dunes, gardens, and suburban parks. It shows remarkable geographic variation: over 50 subspecies have been described, ranging from tiny pale desert birds to large dark subspecies in the Pacific Northwest. Many northern populations migrate south in winter, while southern and coastal populations are permanent residents.

DIET & FEEDING

- White millet — the most attractive seed at feeders
- Black-oil sunflower seeds and cracked corn
- Weed seeds: crabgrass, ragweed, smartweed, knotweed
- Insects and spiders — critical during breeding season
- Wild berries and small fruits in autumn
- Primarily forages on the ground or in low vegetation

BEHAVIOR & SOCIAL LIFE

Song Sparrows are loosely territorial and moderately social. Males defend breeding territories through song, often countersinging with neighboring males in elaborate vocal exchanges. Outside the breeding season they become more gregarious, joining mixed sparrow flocks. They forage primarily on the ground, scratching through leaf litter with a characteristic double-scratch hop. When startled, they fly low and briefly pump their rounded tail — a distinctive behavior that helps identify them in flight. Song Sparrows show strong site fidelity, returning to the same territories year after year.

NESTING & BREEDING

Song Sparrows are among the earliest breeders in North America, with males singing on territory as early as late January in the South. The female builds a well-hidden cup nest of grasses, weeds, and bark strips, placed on the ground or low in dense vegetation. Clutches of 3–5 pale greenish eggs with brown speckles are incubated by the female for 12–14 days. Both parents feed nestlings, which fledge at 10–12 days. Song Sparrows are frequent hosts of Brown-headed Cowbird eggs — a brood parasite that lays its eggs in other birds' nests. Pairs raise 2–3 broods per season.

SONG & COMMUNICATION

The Song Sparrow earns its name through one of the most complex and beautiful songs of any North American sparrow. Each male learns a repertoire of 8–20 distinct song types and switches between them throughout the day. Songs typically begin with 2–3 short notes, followed by a trill and then a complex jumble of notes. The song is often described as beginning with "Maids! Maids! Maids! Pick up your teakettle-ettle-ettle." Each male's songs are slightly different, allowing individual recognition by neighbors.

CONSERVATION STATUS

IUCN Red List:	Least Concern	Population trend:	Stable
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The Song Sparrow population is estimated at 130 million individuals and remains broadly stable, though some regional populations have declined due to wetland loss and habitat fragmentation. It is one of the most studied birds in the world — research on Song Sparrow populations on Mandarte Island, British Columbia has been ongoing since 1975.

FEEDER TIPS

- Scatter white millet on the ground or a low platform feeder
- Dense shrubs and brush piles near feeders provide essential cover
- Song Sparrows often arrive at feeders early in the morning
- A shallow ground-level birdbath is strongly attractive to them
- Leave native grasses and weedy patches in the garden for natural foraging

COLORING GUIDE

ZONE	RECOMMENDED COLOR
Crown	Brown with gray central stripe
Back & wings	Brown, heavily streaked dark
Breast & sides	White with bold brown streaks
Central breast spot	Dark brown — diagnostic dot
Supercilium (eyebrow)	Gray-white stripe above eye
Bill	Short, conical, pale gray

COLOR ME!

Song Sparrow • *Melospiza melodia* • Module 11

